

Final Data

August 6, 2024

1 An Independent Validation of the Final Data

We ultimately found a slight difference in the dataset from this code and realized that data_formatting_pt2_bb.ipynb created the correct dataset to use.

```
[ ]: # Data analysis
import pandas as pd
import numpy as np

# Plotting and Correlation Maths
import seaborn as sns
import scipy as sci

# Simple model development
import sklearn as sk
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.decomposition import TruncatedSVD, PCA
from sklearn.linear_model import LogisticRegression
from sklearn.neural_network import MLPClassifier
from sklearn.ensemble import RandomForestClassifier

import matplotlib.pyplot as plt
import plotly as px

import os
```

```
[ ]: file_path = '/Users/hunterberberich/Downloads/cleaned_data (1).csv'
file_path2 = '/Users/hunterberberich/Desktop/Data/merged_df.csv'
cleaned_df = pd.read_csv(file_path)
merged_df = pd.read_csv(file_path2)
```

```
[ ]: #Unique Key for merge
cleaned_df['unique_key'] = cleaned_df['tempmax'].astype(str) + '_' +
    ↪cleaned_df['tempmin'].astype(str)
merged_df['unique_key'] = merged_df['tempmax'].astype(str) + '_' +
    ↪merged_df['tempmin'].astype(str)
merged_df_unique = merged_df.drop_duplicates(subset='unique_key', keep='first')
```

```

survive_dict = dict(zip(merged_df_unique['unique_key'],
    ↪merged_df_unique['survive']))

# 'survive' column to cleaned_df by matching 'unique_key'
cleaned_df['survive'] = cleaned_df['unique_key'].map(survive_dict)

survive_df = cleaned_df.copy()

print("Survive DataFrame Head:\n", survive_df.head())

# Check data for accuracy
print(f"Original cleaned_df shape: {cleaned_df.shape}")
print(f"New survive_df shape: {survive_df.shape}")

missing_values = survive_df['survive'].isnull().sum()
print(f"Number of missing values in 'survive' column: {missing_values}")

spot_check_indices = [0, 10, 50] # Example indices to check
print("Spot check specific rows in survive_df:")
print(survive_df.loc[spot_check_indices, ['tempmax', 'tempmin', 'survive',
    ↪'unique_key']])
print("Corresponding rows in merged_df:")
print(merged_df.loc[merged_df['unique_key'].isin(survive_df.
    ↪loc[spot_check_indices, 'unique_key']), ['tempmax', 'tempmin', 'survive',
    ↪'unique_key']])

missing_keys = survive_df[~survive_df['unique_key'].
    ↪isin(merged_df_unique['unique_key'])]
print(f"Number of keys in survive_df not found in merged_df: {missing_keys.
    ↪shape[0]}")

survive_df = survive_df.drop(columns=['unique_key'])

result_file_path = '/Users/hunterberberich/Downloads/cleaned_data_with_survive2.
    ↪csv'
survive_df.to_csv(result_file_path, index=False)

```

Survive DataFrame Head:

	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	\
0	26.2	-4.2	10.1	18.1	-16.6	0.6	-3.8	
1	35.8	28.0	32.5	34.7	24.1	26.8	26.6	
2	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	
3	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	
4	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	

	humidity	precip	precipcover	...	preciptype_freezingrain,snow,ice	\
0	57.2	0.0	0.0	...		0
1	78.9	0.0	0.0	...		0
2	68.0	0.0	0.0	...		0
3	68.0	0.0	0.0	...		0
4	68.0	0.0	0.0	...		0

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0		0
1	1	0		0
2	1	0		0
3	1	0		0
4	1	0		0

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	

	preciptype_rain,snow,ice	preciptype_snow	unique_key	survive
0	0	0	26.2_-4.2	Y
1	0	0	35.8_28.0	Y
2	0	0	15.3_2.5	Y
3	0	0	15.3_2.5	Y
4	0	0	15.3_2.5	Y

[5 rows x 28 columns]

Original cleaned_df shape: (3684, 28)

New survive_df shape: (3684, 28)

Number of missing values in 'survive' column: 0

Spot check specific rows in survive_df:

	tempmax	tempmin	survive	unique_key
0	26.2	-4.2	Y	26.2_-4.2
10	55.8	33.0	N	55.8_33.0
50	41.1	27.2	Y	41.1_27.2

Corresponding rows in merged_df:

	tempmax	tempmin	survive	unique_key
0	26.2	-4.2	Y	26.2_-4.2
11	55.8	33.0	N	55.8_33.0
12	55.8	33.0	U	55.8_33.0
13	55.8	33.0	N	55.8_33.0
14	55.8	33.0	N	55.8_33.0
15	55.8	33.0	N	55.8_33.0
59	41.1	27.2	Y	41.1_27.2
60	41.1	27.2	N	41.1_27.2

```

61      41.1      27.2      Y  41.1_27.2
62      41.1      27.2      N  41.1_27.2
Number of keys in survive_df not found in merged_df: 0

```

```

[ ]: # Remove 'U' to make 'survive' Binary
survive_df['survive'] = survive_df['survive'].replace('U', pd.NA)
survive_df['survive'] = survive_df['survive'].map({'Y': 1, 'N': 0})

survive_df

```

```

[ ]:      tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew \
0         26.2     -4.2  10.1         18.1         -16.6         0.6 -3.8
1         35.8     28.0  32.5         34.7         24.1        26.8 26.6
2         15.3      2.5   9.2         14.6         -6.0         3.5  0.5
3         15.3      2.5   9.2         14.6         -6.0         3.5  0.5
4         15.3      2.5   9.2         14.6         -6.0         3.5  0.5
...
3679      68.8      58.9  61.6         68.8         58.9        61.6 56.7
3680      59.4      46.8  50.8         59.4         39.8        46.5 46.3
3681      59.2      44.5  51.1         59.2         37.4        47.9 33.0
3682      81.1      45.0  64.1         79.9         39.9        62.9 39.5
3683      81.1      45.0  64.1         79.9         39.9        62.9 39.5

```

```

      humidity  precip  precipcover  ...  precip_type_freezingrain,snow \
0         57.2    0.000         0.00  ...                                0
1         78.9    0.000         0.00  ...                                0
2         68.0    0.000         0.00  ...                                0
3         68.0    0.000         0.00  ...                                0
4         68.0    0.000         0.00  ...                                0
...
3679      84.8    1.526         83.33  ...                                0
3680      84.5    0.266         83.33  ...                                0
3681      51.5    0.019         12.50  ...                                0
3682      42.6    0.001          4.17  ...                                0
3683      42.6    0.001          4.17  ...                                0

```

```

      precip_type_freezingrain,snow,ice  precip_type_none  precip_type_rain \
0                                     0                   1                   0
1                                     0                   1                   0
2                                     0                   1                   0
3                                     0                   1                   0
4                                     0                   1                   0
...
3679                                     0                   0                   1
3680                                     0                   0                   1
3681                                     0                   0                   1
3682                                     0                   0                   1

```

3683		0	0	1
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	preciptype_rain,freezingrain	preciptype_rain,freezingrain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
3679	0	0	
3680	0	0	
3681	0	0	
3682	0	0	
3683	0	0	

	preciptype_rain,snow	preciptype_rain,snow,ice	preciptype_snow	survive
0	0	0	0	1.0
1	0	0	0	1.0
2	0	0	0	1.0
3	0	0	0	1.0
4	0	0	0	1.0
...	...	...	...	
3679	0	0	0	NaN
3680	0	0	0	1.0
3681	0	0	0	1.0
3682	0	0	0	1.0
3683	0	0	0	1.0

[3684 rows x 27 columns]

```
[ ]: survive_df = survive_df.dropna(subset=['survive'])
```

```
survive_df
```

```
[ ]:      tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew  \
0         26.2     -4.2  10.1         18.1         -16.6      0.6  -3.8
1         35.8     28.0  32.5         34.7          24.1     26.8  26.6
2         15.3      2.5   9.2          14.6          -6.0      3.5   0.5
3         15.3      2.5   9.2          14.6          -6.0      3.5   0.5
4         15.3      2.5   9.2          14.6          -6.0      3.5   0.5
...      ...      ...   ...      ...      ...      ...
3678      65.1     57.1  60.6         65.1         57.1     60.6  54.5
3680      59.4     46.8  50.8         59.4         39.8     46.5  46.3
3681      59.2     44.5  51.1         59.2         37.4     47.9  33.0
3682      81.1     45.0  64.1         79.9         39.9     62.9  39.5
3683      81.1     45.0  64.1         79.9         39.9     62.9  39.5
```

	humidity	precip	precipcover	...	preciptype_freezingrain,snow	\
0	57.2	0.000	0.00	...	0	
1	78.9	0.000	0.00	...	0	
2	68.0	0.000	0.00	...	0	
3	68.0	0.000	0.00	...	0	
4	68.0	0.000	0.00	...	0	
...	...	...	...	...	...	
3678	80.9	0.179	62.50	...	0	
3680	84.5	0.266	83.33	...	0	
3681	51.5	0.019	12.50	...	0	
3682	42.6	0.001	4.17	...	0	
3683	42.6	0.001	4.17	...	0	

	preciptype_freezingrain,snow,ice	preciptype_none	preciptype_rain	\
0	0	1	0	
1	0	1	0	
2	0	1	0	
3	0	1	0	
4	0	1	0	
...	...	...	...	
3678	0	0	1	
3680	0	0	1	
3681	0	0	1	
3682	0	0	1	
3683	0	0	1	

	preciptype_rain,freezingrain	preciptype_rain,freezingrain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
3678	0	0	
3680	0	0	
3681	0	0	
3682	0	0	
3683	0	0	

	preciptype_rain,snow	preciptype_rain,snow,ice	preciptype_snow	survive
0	0	0	0	1.0
1	0	0	0	1.0
2	0	0	0	1.0
3	0	0	0	1.0
4	0	0	0	1.0
...	...	...	...	
3678	0	0	0	1.0

3680	0	0	0	1.0
3681	0	0	0	1.0
3682	0	0	0	1.0
3683	0	0	0	1.0

[3440 rows x 27 columns]

```
[ ]: result_file_path = '/Users/hunterberberich/Downloads/survive_cleaned.csv'  
survive_df.to_csv(result_file_path, index=False)
```

```
[ ]:
```